

# Stability Analysis and Neural Network Modelling of Clustered Large-Scale Brain Dynamics

Master Thesis Defence

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Universitat Politècnica de Catalunya – Facultat de Matemàtiques i Estadística, June 19, 2026

# Outline

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Introduction

Large-Scale Brain Model

Structural Connectivity Clustering

Linear Stability Analysis of Homogeneous Solutions

DeepONet Approximation for Phase-Amplitude Dynamics

Conclusions

Questions

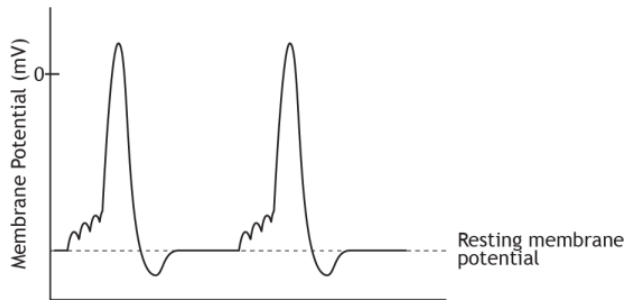
# Introduction

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## Introduction: Brain Modelling

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- Neuronal activity in the brain exhibits both **oscillatory** and **resting-state dynamics**.
- One possible approach to describing these dynamics is through **parameter-dependent dynamical systems**.



**Figure 1:** **Oscillatory** behaviour of neuronal membrane potentials. Extracted from [1].

## Introduction: Brain Modelling

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Comes with several challenges

- **Long-range interactions** between different brain regions
- **High dimensionality**, leading to limited interpretability and high computational cost
- Multiple dynamical behaviours and **stability regimes**

## Introduction: Objectives

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Comes with several challenges → **Objective: address them**

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- Multiple dynamical behaviours and stability regimes → **stability analysis**

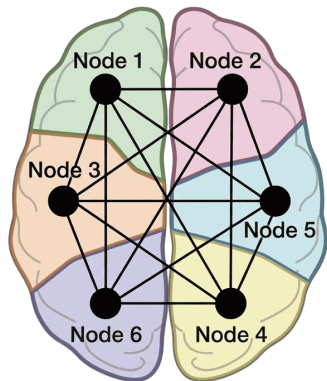
Continuation of the work by **Delicado et al.** (2026) [2].

# Large-Scale Brain Model

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## Large-Scale Brain Model: Structural Connectivity

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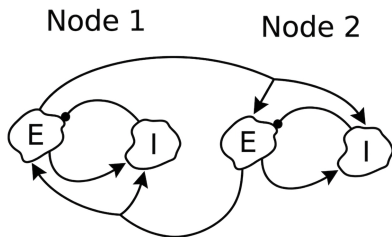


**Figure 2:** Schematic of the large-scale structural connectivity.

- Structural connectivity graph.
- Nodes correspond to brain regions; edges represent long-range interactions.
- **Connectivity matrix:**  $(w_{ij}) \in \mathbb{R}^{N \times N}$ .
- Derived from diffusion MRI tractography data and mapped onto the AAL90 atlas ( $N = 90$ ) (Deco et al. [3]).
- Clustering techniques can be used to construct reduced-order models with fewer nodes.

## Large-Scale Brain Model: NG-NMM Node Dynamics

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**Figure 3:** Schematic of excitatory–inhibitory interactions within and between nodes.

- At each node, interacting excitatory (E) and inhibitory (I) subpopulations are represented by ensembles of all-to-all coupled QIF neurons.
- Local dynamics are modelled by a **next-generation neural mass model (NG-NMM)**.
- The model describes the time evolution of variables characterizing neuronal activity:
  1. Mean firing rates,  $r_E$  and  $r_I$ ;
  2. Mean membrane potentials,  $v_E$  and  $v_I$ ;
  3. Synaptic activities,  $s_E$  and  $s_I$ .

## Large-Scale Brain Model: System of Differential Equations

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System of  $6N$  autonomous ordinary differential equations. For each node  $j \in \{1, \dots, N\}$ :

$$\begin{aligned} (E_j) \quad & \begin{cases} \tau_E \dot{r}_{E,j} = \frac{\Delta_E}{\pi \tau_E} + 2r_{E,j}v_{E,j}, \\ \tau_E \dot{v}_{E,j} = \bar{\eta}_E + v_{E,j}^2 - (\tau_E \pi r_{E,j})^2 + I_{\text{ext}}^E + \tau_E (J_{EE}s_{E,j} - J_{EI}s_{I,j} + I_j), \\ \tau_{s_E} \dot{s}_{E,j} = -s_{E,j} + r_{E,j}. \end{cases} \\ (I_j) \quad & \begin{cases} \tau_I \dot{r}_{I,j} = \frac{\Delta_I}{\pi \tau_I} + 2r_{I,j}v_{I,j}, \\ \tau_I \dot{v}_{I,j} = \bar{\eta}_I + v_{I,j}^2 - (\tau_I \pi r_{I,j})^2 + I_{\text{ext}}^I + \tau_I (J_{IE}s_{E,j} - J_{II}s_{I,j} + I_j), \\ \tau_{s_I} \dot{s}_{I,j} = -s_{I,j} + r_{I,j}. \end{cases} \end{aligned} \tag{1}$$

## Large-Scale Brain Model: System of Differential Equations

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$$I_j = \varepsilon \sum_{k=1}^N \tilde{w}_{jk} s_{E,k}, \quad \text{s.t.} \quad \sum_{k=1}^N \tilde{w}_{jk} = 1. \quad (2)$$

# Structural Connectivity Clustering

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# Structural Connectivity Clustering: Introduction

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**Goal:** Reduce the dimensionality of the structural connectivity via graph clustering:

$$\underbrace{W}_{\text{empirical}} \in \mathbb{R}^{90 \times 90} \longrightarrow \underbrace{w}_{\text{clustered}} \in \mathbb{R}^{k \times k}, \quad k \ll 90.$$

**Method: Spectral clustering:**

- Simple and interpretable.
- Explores the dynamics preservation in the following way:

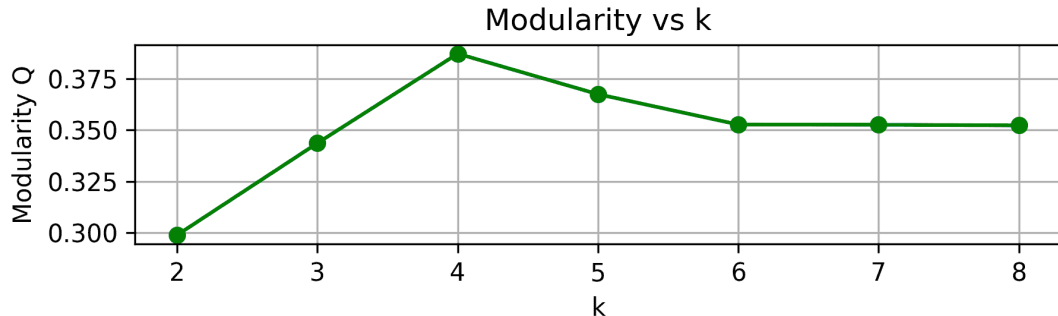
Preservation of the dominant  
eigenvalues of the  
connectivity matrix.

$\xrightarrow{?}$

Preservation of system stability  
characteristics.

## Structural Connectivity Clustering: Quality metric

Apply the spectral clustering method for different numbers of clusters  $k$  and assess the quality:



**Figure 4:** Clustering  $Q$ -modularity metric for a low number of clusters; larger numbers of clusters result in lower metric values. An underlying community structure is observed for the [4-clustering](#).

# Linear Stability Analysis of Homogeneous Solutions

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## Linear Stability Analysis: Introduction

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**Goal:** Determine the stability of homogeneous solutions under transverse perturbations.

- Homogeneous solutions: equilibrium points (EPs) and periodic orbits (POs).
- Relationship between the **Jacobian eigenvalues** and the **structural connectivity spectrum**.
- Dependence on the input parameter  $I_{ext}^E$  and the coupling strength  $\epsilon$ .

## Linear Stability Analysis: Homogeneous Invariant Manifold

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Let  $\mathbf{x}_j = (r_{E,j}, v_{E,j}, s_{E,j}, r_{I,j}, v_{I,j}, s_{I,j}) \equiv$  variables describing the dynamics of node  $j$ .

Impose:  $\mathbf{x}_j = \mathbf{x} = (r_E, v_E, s_E, r_I, v_I, s_I)$  for all  $j = 1, \dots, N$ .

Then:

$$l_j = \epsilon \sum_{k=1}^N \tilde{w}_{jk} s_{E,k} = \epsilon s_E \sum_{k=1}^N \tilde{w}_{jk} = \epsilon s_E, \quad (3)$$

## Linear Stability Analysis: Homogeneous Invariant Manifold

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Full network system of  $6N$  equations reduces to 6 equations, independent of the connectivity structure:

$$\begin{aligned} (E) \quad & \begin{cases} \tau_E \dot{r}_E = \frac{\Delta_E}{\pi \tau_E} + 2r_E v_E, \\ \tau_E \dot{v}_E = \bar{\eta}_E + v_E^2 - (\tau_E \pi r_E)^2 + I_{\text{ext}}^E + \tau_E ((J_{EE} + \epsilon)s_E - J_{EI}s_I), \\ \tau_{s_E} \dot{s}_E = -s_E + r_E. \end{cases} \\ (I) \quad & \begin{cases} \tau_I \dot{r}_I = \frac{\Delta_I}{\pi \tau_I} + 2r_I v_I, \\ \tau_I \dot{v}_I = \bar{\eta}_I + v_I^2 - (\tau_I \pi r_I)^2 + I_{\text{ext}}^I + \tau_I ((J_{IE} + \epsilon)s_E - J_{II}s_I), \\ \tau_{s_I} \dot{s}_I = -s_I + r_I. \end{cases} \end{aligned} \tag{4}$$

## Linear Stability Analysis: Perturbation Projection

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We study the linear stability of homogeneous solutions  $x^{(0)}$  through the family of Jacobians depending on the structural connectivity eigenmodes:

$$\mathcal{J}(x^{(0)}, \Lambda_\alpha) := J(x^{(0)}) + \varepsilon \Lambda_\alpha K, \quad \alpha = 1, \dots, N$$

## Linear Stability Analysis: Perturbation Projection

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Let  $x^{(0)}$  be an equilibrium point (EP), and let  $\mu_j^{(\alpha)}$  denote the eigenvalues of  $\mathcal{J}(x^{(0)}, \Lambda_\alpha)$ .

**Dispersion relation:**

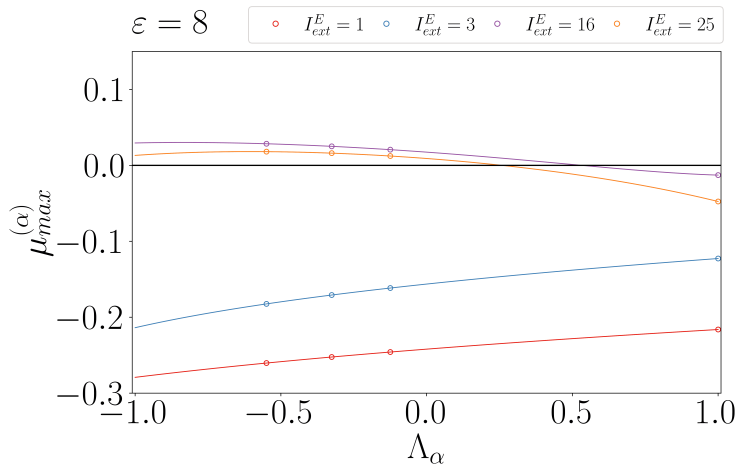
$$\mu_{\max}^{(\alpha)} = \max_{1 \leq j \leq 6} \Re(\mu_j^{(\alpha)}) = \mathcal{F}(\Lambda_\alpha).$$

- $\mu_{\max}^{(\alpha)} > 0 \implies$  instability along the  $\alpha$ -th eigenmode.
- $\mu_{\max}^{(\alpha)} < 0 \implies$  stability along the  $\alpha$ -th eigenmode.

For  $x^{(0)}$  a periodic orbit (PO), the analogous mapping is known as the **master stability function**, where  $\mu_j^{(\alpha)}$  are the Floquet exponents of  $\mathcal{J}(x^{(0)}, \Lambda_\alpha)$ .

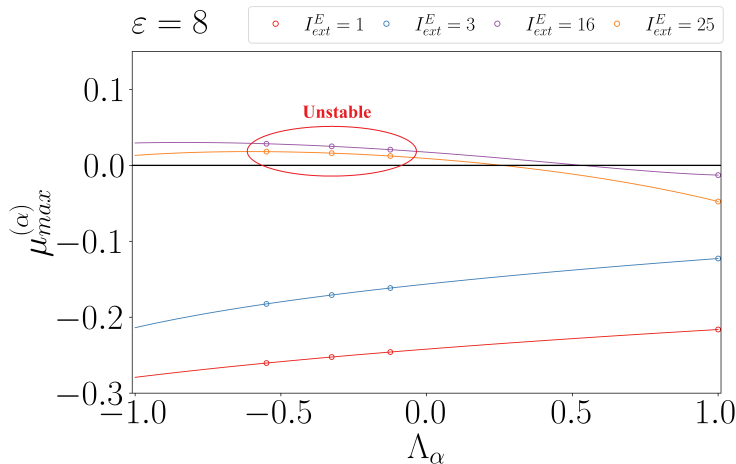


## Linear Stability Analysis: Dispersion Relation



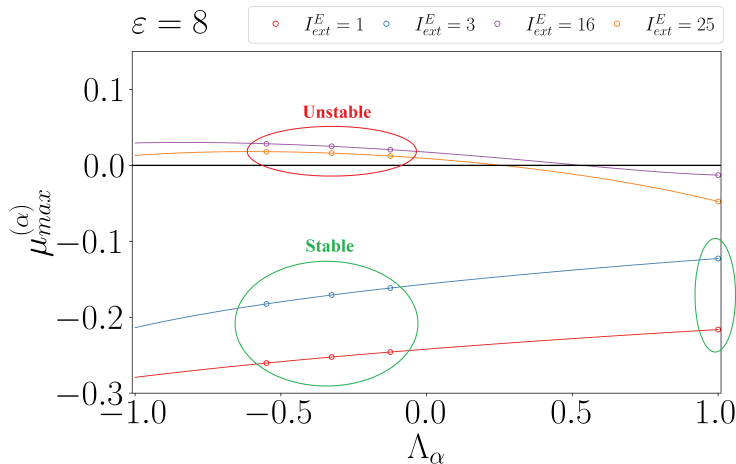
**Figure 5:** Dispersion relations for  $\varepsilon = 8$  and different values of  $I_{ext}^E$ . Generated using adapted code from [2].

## Linear Stability Analysis: Dispersion Relation



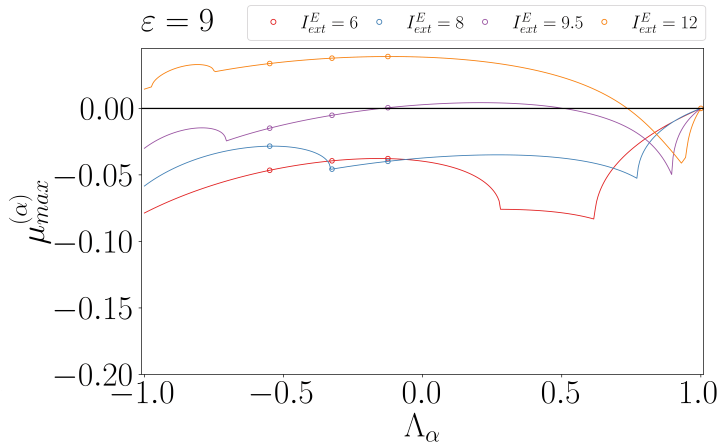
**Figure 6:** Dispersion relations for  $\varepsilon = 8$  and different values of  $I_{ext}^E$ . Generated using adapted code from [2].

## Linear Stability Analysis: Dispersion Relation



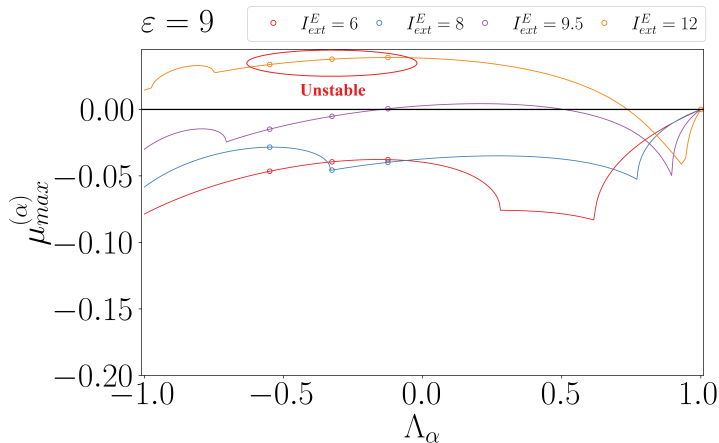
**Figure 7:** Dispersion relations for  $\varepsilon = 8$  and different values of  $I_{ext}^E$ . Generated using adapted code from [2].

## Linear Stability Analysis: Master Stability Function



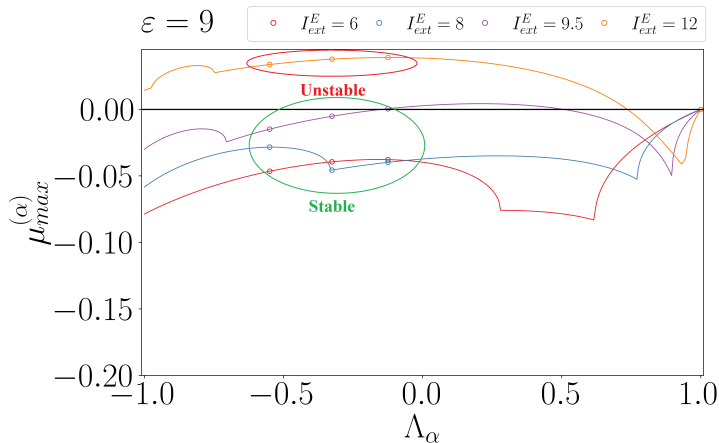
**Figure 8:** Master stability functions for  $\varepsilon = 9$  and different values of  $I_{ext}^E$ . Generated using adapted code from [2].

## Linear Stability Analysis: Master Stability Function



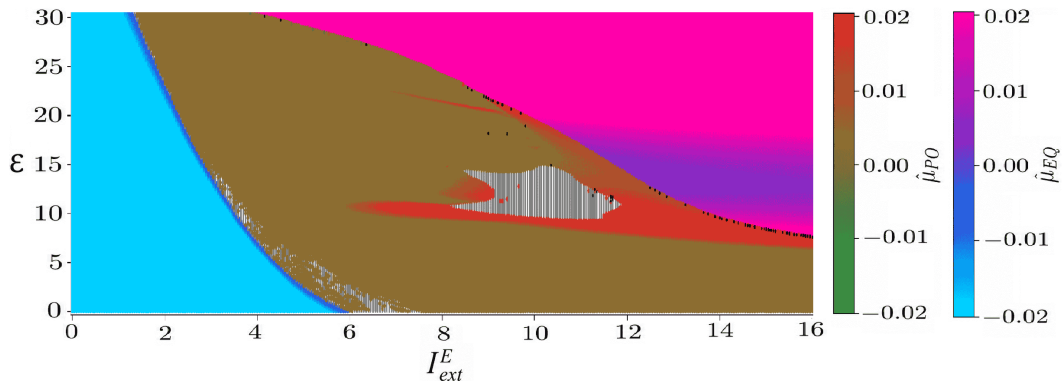
**Figure 9:** Master stability functions for  $\varepsilon = 9$  and different values of  $I_{ext}^E$ . Generated using adapted code from [2].

## Linear Stability Analysis: Master Stability Function



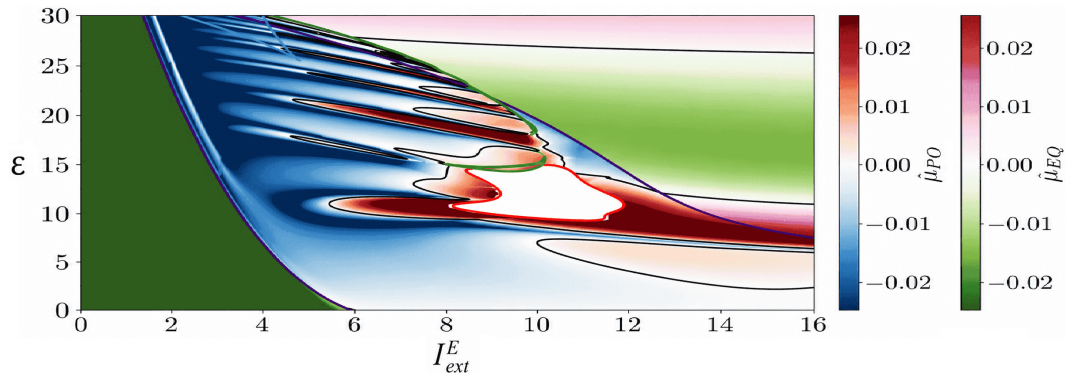
**Figure 10:** Master stability functions for  $\varepsilon = 9$  and different values of  $I_{ext}^E$ . Generated using adapted code from [2].

## Linear Stability Analysis: Bifurcation Diagram (Clustered)



**Figure 11:** Bifurcation diagram showing the largest exponent over the eigenmodes of the clustered structural connectivity  $\tilde{w}$ .

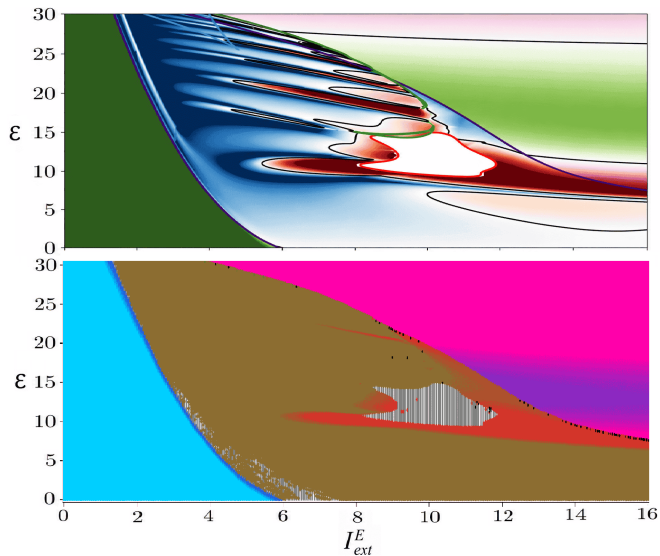
## Linear Stability Analysis: Bifurcation Diagram (Original)



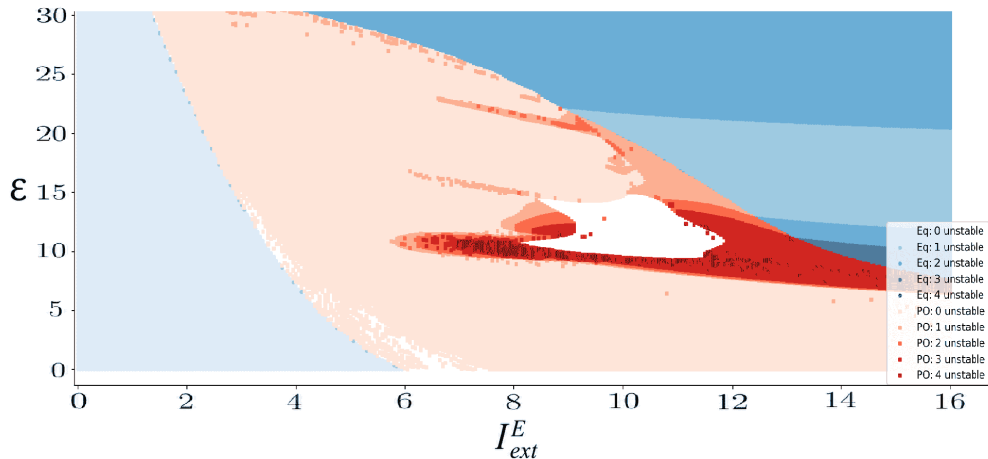
**Figure 12:** Bifurcation diagram showing the largest exponent over the eigenmodes of the original structural connectivity  $\tilde{W}$ , excluding the exponent corresponding to the eigenmode  $\Lambda_\alpha = 1$ . Extracted from [2].



# Linear Stability Analysis: Bifurcation Diagram (Comparison)

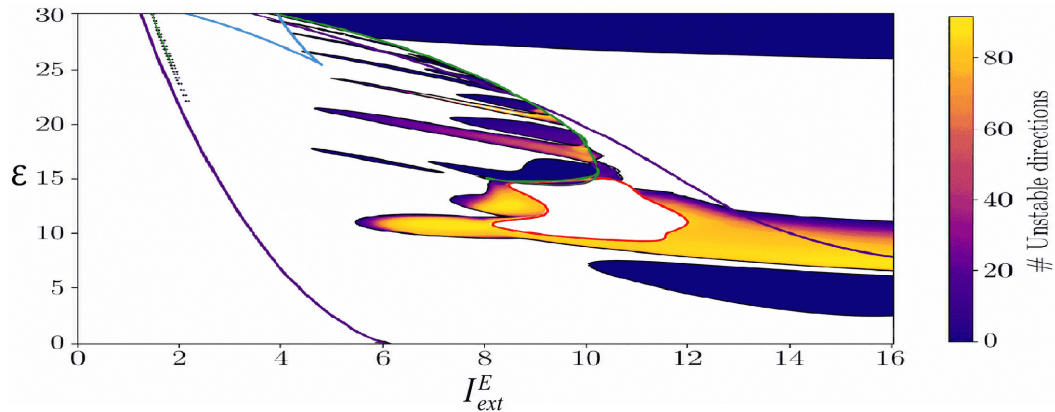


## Linear Stability Analysis: Bifurcation Diagram (Clustered)



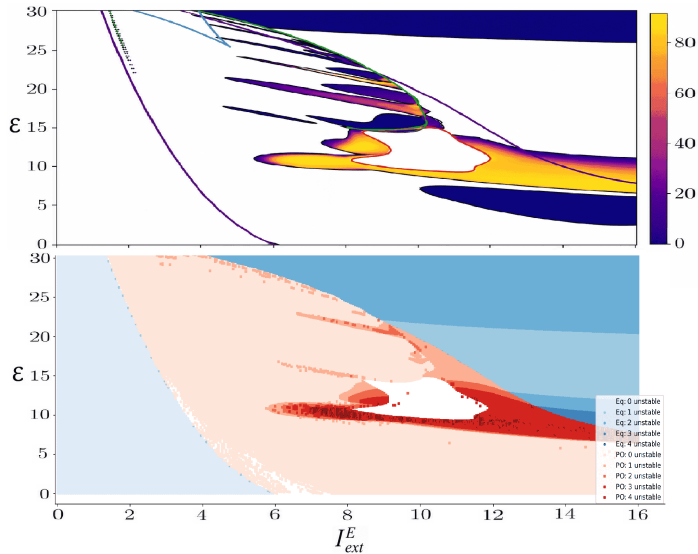
**Figure 13:** Bifurcation diagram complemented with the corresponding number of unstable directions (i.e., positive exponents) across the eigenmodes of the clustered structural connectivity  $\tilde{w}$ . 30/56

## Linear Stability Analysis: Bifurcation Diagram (Original)



**Figure 14:** Bifurcation diagram showing the number of unstable directions across the eigenmodes of the original structural connectivity  $\tilde{W}$ . Extracted from [2].

# Linear Stability Analysis: Bifurcation Diagram (Comparison)



## Linear Stability Analysis: Conclusions

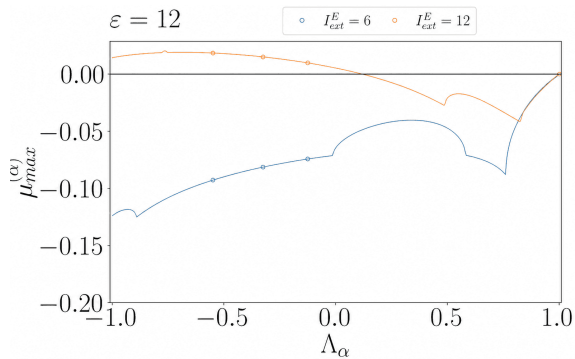
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- Spectral clustering reduces the empirical structural connectivity to a low-dimensional network with four nodes.
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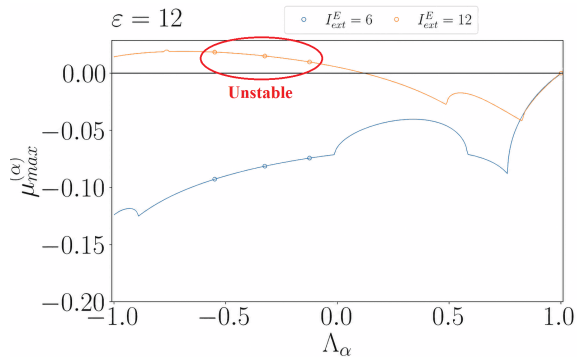
# DeepONet Approximation for Phase-Amplitude Dynamics

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# DeepONet Approximation: Introduction

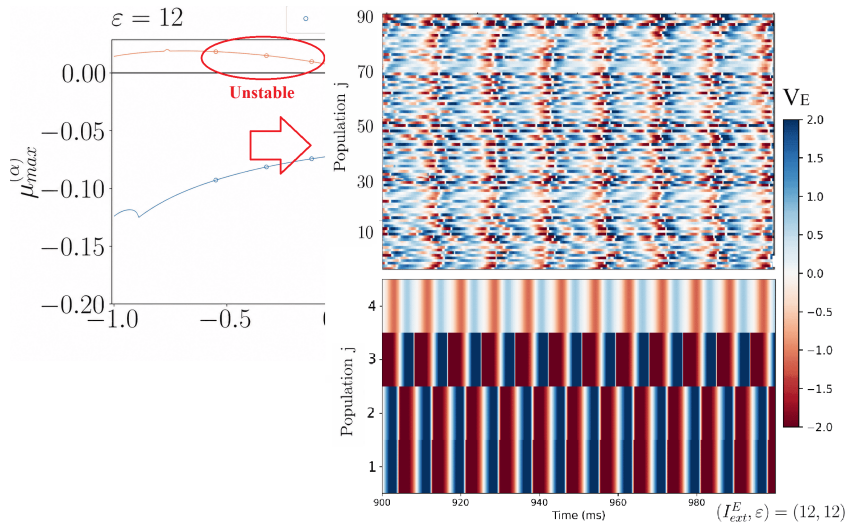


# DeepONet Approximation: Introduction

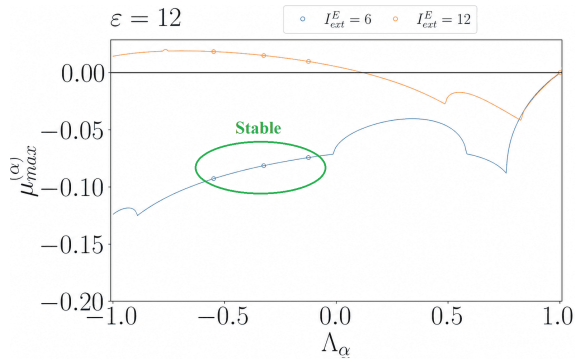




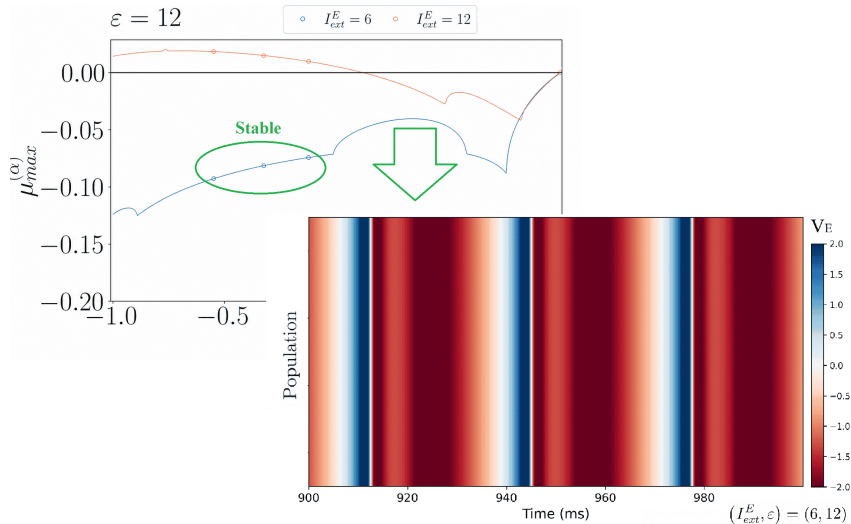
# DeepONet Approximation: Introduction



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## DeepONet Approximation: Introduction

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What we will study next:

- Explore the dynamics **around** **stable homogeneous periodic orbits** using a **phase–amplitude model**.
- Predict these dynamics using a **DeepONet** approximation [5].

## DeepONet Approximation: Introduction

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**Goal:** Introduce DeepONets for predicting general solutions to parameter-dependent initial value problems and apply them to approximate the solution of the phase–amplitude model.

- Present the phase–amplitude model, a low-dimensional system describing the local dynamics around a stable homogeneous periodic orbit.
- Describe the DeepONet architecture and adapt it to the specific prediction problem.
- Develop a software implementation of the methodology and demonstrate its predictive accuracy.

## DeepONet Approximation: Phase-Amplitude Coupling Dynamics

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Localise a stable homogeneous periodic orbit for fixed  $(I_{\text{ext}}^E, \varepsilon = 0)$  and compute:

- its period,  $T$ ;
- its largest non-zero Floquet exponent,  $\lambda < 0$ .

For illustration, we take  $I_{\text{ext}}^E = 10$ .

## DeepONet Approximation: Phase-Amplitude Coupling Dynamics

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Reduced dynamics for  $\varepsilon > 0$  restricted to the attracting manifold associated with  $\lambda$  in a neighbourhood of the periodic orbit, expressed in phase–amplitude coordinates [4]:

$$\begin{cases} \dot{\theta} = 1 + \varepsilon Z(\theta, \sigma) K^{SE}(\theta, \sigma), & \theta \in [0, T), \\ \dot{\sigma} = \lambda \sigma + \varepsilon I(\theta, \sigma) K^{SE}(\theta, \sigma), & \sigma \in \mathbb{R}. \end{cases} \quad (5)$$

Our objective is to approximate the general solution

$$\phi(\varepsilon, (\theta_0, \sigma_0))(t) = (\theta(\varepsilon, \theta_0, \sigma_0)(t), \sigma(\varepsilon, \theta_0, \sigma_0)(t)) \quad (6)$$

using the **DeepONet neural network architecture**.

## DeepONet Approximation: Architecture

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Consider the explicit solution of the uncoupled system:

$$\phi(\varepsilon = 0, \theta_0, \sigma_0)(t) \equiv \phi_b(\theta_0, \sigma_0)(t) = (t + \theta_0, \sigma_0 \exp(\lambda t)). \quad (7)$$

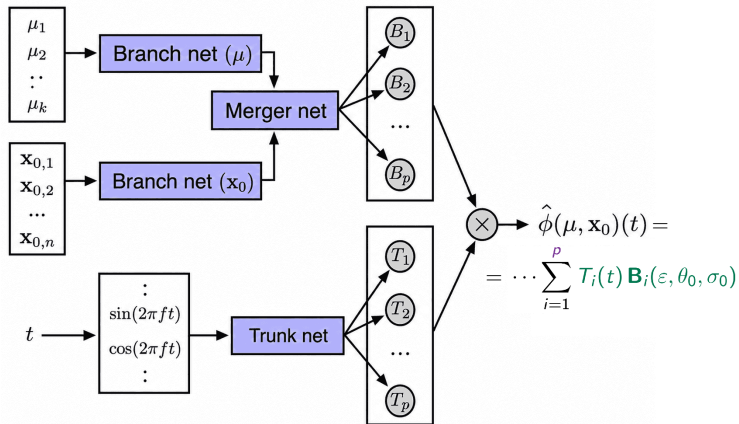
We then propose the approximation:

$$\hat{\phi}(\varepsilon, \theta_0, \sigma_0)(t) = \phi_b(\theta_0, \sigma_0)(t) + \varepsilon(1 - \exp(-t)) \sum_{i=1}^p T_i(t) \mathbf{B}_i(\varepsilon, \theta_0, \sigma_0). \quad (8)$$

Here,  $T_i$  and  $\mathbf{B}_i$  denote the outputs of the **trunk** and **branch** networks, respectively, in a DeepONet architecture with basis size  $p$ .



## DeepONet Approximation: Architecture



**Figure 16:** Illustration of the adapted DeepONet architecture for IVP oscillatory solution approximation.

## DeepONet Approximation: Training

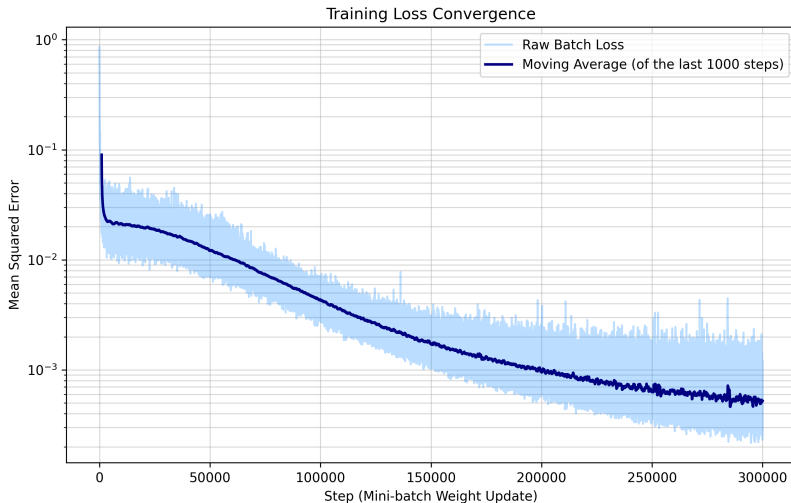
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To train the parameters of the DeepONet, consider:

$$L_{\text{superv}} = \left\| \hat{\theta}(\varepsilon, \theta_0)(t) - \theta(\varepsilon, \theta_0)(t) \right\|^2 + \beta \left\| \hat{\sigma}(\varepsilon, \sigma_0)(t) - \sigma(\varepsilon, \sigma_0)(t) \right\|^2, \quad (9)$$

with **reference trajectories** obtained numerically by sampling the parameter values, initial conditions, and evaluation times.

# DeepONet Approximation: Training



**Figure 17:** Loss convergence during the training process on a logarithmic scale.

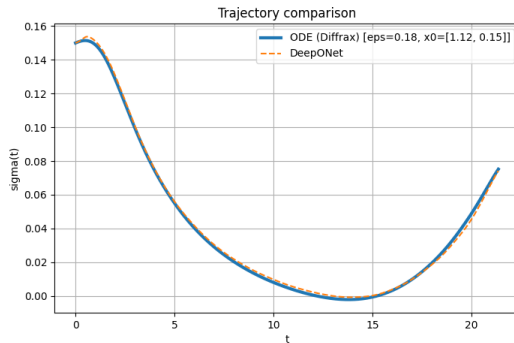
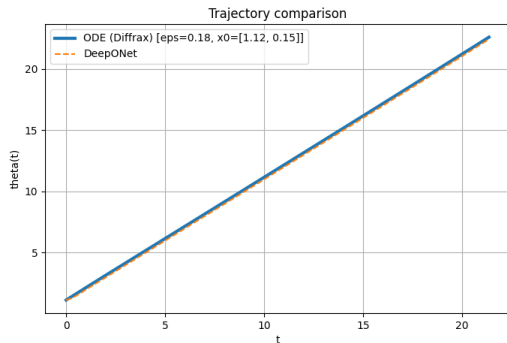
## DeepONet Approximation: Prediction

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Once the DeepONet has been trained, predictions are obtained by evaluating the network as follows:

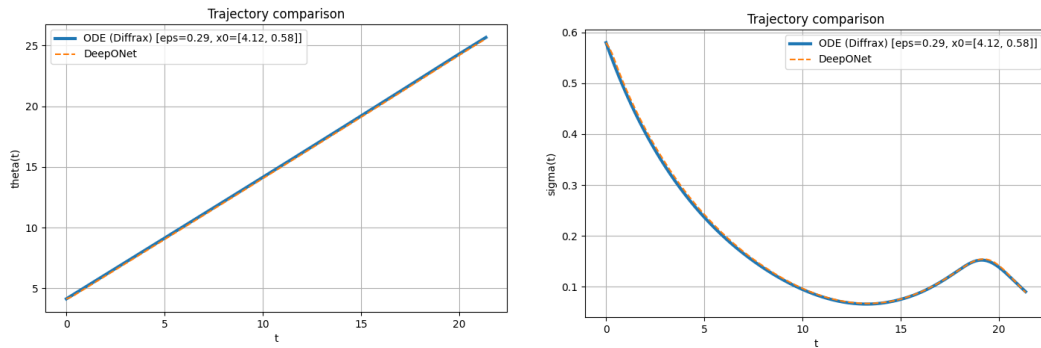
$$\underbrace{(t, \varepsilon, \theta_0, \sigma_0)}_{\text{input}} \xrightarrow{\text{DeepONet}} \underbrace{\hat{\phi}(\varepsilon, \theta_0, \sigma_0)(t)}_{\text{output}}.$$

# DeepONet Approximation: Prediction



**Figure 18:** DeepONet prediction of the phase-amplitude system for  $\varepsilon = 0.18$ .

# DeepONet Approximation: Prediction



**Figure 19:** DeepONet prediction of the phase-amplitude system for  $\varepsilon = 0.29$ .

# DeepONet Approximation: Inverse Problem

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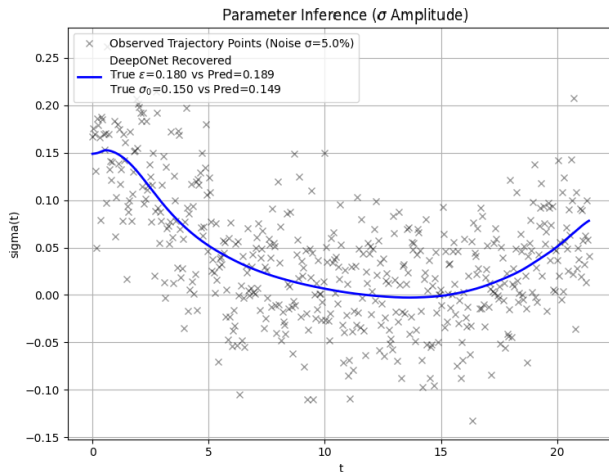
**Inverse problem (IP):**

$$\underbrace{((t_1, \dots, t_N), (\tilde{\phi}(t_1), \dots, \tilde{\phi}(t_N)))}_{\text{input}} \xrightarrow{\text{IP prediction}} \underbrace{(\varepsilon^*, \theta_0^*, \sigma_0^*)}_{\text{output}}.$$

**Goal:** Determine the parameters  $(\varepsilon^*, \theta_0^*, \sigma_0^*)$  such that  $\hat{\phi}(\varepsilon^*, \theta_0^*, \sigma_0^*)(t)$  provides the best fit to the observed data.

**Note:** The observations  $\tilde{\phi}(t_i)$  may be noisy!

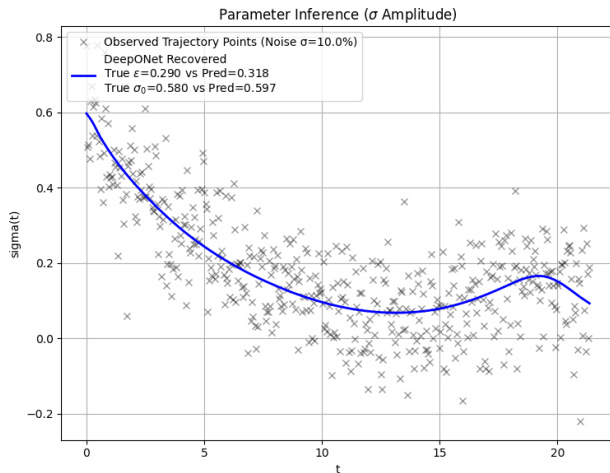
# DeepONet Approximation: Inverse Problem



**Figure 20:** DeepONet-predicted trajectory obtained from the inverse problem.



# DeepONet Approximation: Inverse Problem



**Figure 21:** DeepONet-predicted trajectories obtained from the inverse problem.

## Conclusions

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## Conclusions

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- Spectral clustering reduces the empirical structural connectivity to a low-dimensional network with four nodes.
- The stability of homogeneous solutions is linked to the spectrum of the clustered connectivity via dispersion relations and the master stability function.
- The clustered model preserves the key stability properties of the original system over most of the explored parameter range.
- DeepONet can be adapted to approximate solutions of parameter-dependent initial value problems.
- DeepONet accurately predicts phase–amplitude dynamics and can recover model parameters from noisy trajectories.

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## References of the Presentation i

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- [1] Henley, Casey L. 'Action Potentials.' <https://openbooks.lib.msu.edu/neuroscience/chapter/action-potentials/>
- [2] Delicado-Moll, Rosa Maria, Gemma Huguet, and Pau Clusella. 'Emergent Spatiotemporal Dynamics in Large-Scale Brain Networks with next Generation Neural Mass Models'. Physica D: Nonlinear Phenomena 493 (September 2026): 135232.
- [3] Deco, Gustavo, Joana Cabral, Mark W. Woolrich, Angus B. A. Stevner, Tim J. Van Hartevelt, and Morten L. Kringelbach. 'Single or Multiple Frequency Generators in On-Going Brain Activity: A Mechanistic Whole-Brain Model of Empirical MEG Data'. NeuroImage 152 (May 2017): 538–50.

- [4] David Reyner-Parra, Alberto Pérez-Cervera, and Gemma Huguet. Efficient numerical algorithms for phase-amplitude reduction on the slow attracting manifold of limit cycles. *Journal of Nonlinear Science*, 36(2), 2026.
- [5] Lu, Lu, Pengzhan Jin, and George Em Karniadakis. 'DeepONet: Learning Nonlinear Operators for Identifying Differential Equations Based on the Universal Approximation Theorem of Operators'. *Nature Machine Intelligence* 3, no. 3 (2021): 218–29.

## Questions

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